shows the annotation interface. We also surveyed the workers about the movies they had watched. We qualified 69 workers who correctly answered at least 80% of the questions and had watched at least ten movies out of the ones sampled for evaluation.

The qualified workers labeled whether the character portrayed the trope on a Likert scale: *no*, *maybe no*, *not sure*, *maybe yes* and *yes*. To reduce the cognitive load on the workers, they annotated the character-trope pairs in batches, each batch containing characters from at most ten movies. Before every batch, we inform the workers about the movies they will encounter to prepare them for the annotation task. After each batch, we estimate the worker's performance by calculating the accuracy of their ratings against the CHATTER labels and record how much time they spent on the task. We dropped workers whose accuracy dropped below 50% or those speeding through the samples, disqualifying them from working on future batches. Of the 69 workers, we disqualified 10 raters and were left with 59 reliable workers.

The annotation task ran for one month between August 11 to September 19 2024. Throughout the task, we maintained a communication channel for the workers in the TurkerNation Slack Workspace, where we responded to any questions the workers had about the task. We pay \$1.5 for each question which turns out to be \$18/hour because the workers spent an average of 5 minutes per question. The entire task cost us about \$10K. We collected three annotations per character-trope pair, totaling 5683 annotations. The Krippendorff inter-rater reliability score is 0.448, indicating moderate agreement.

3.2 CHATTEREVAL

We aggregate the annotations to build the CHATTEREVAL dataset. The character-trope pairs do not all show clear agreement. Character attribution, like sentiment analysis, is a subjective task, and whether a character portrays some trope is perceived differently by people. We need to assign a definite binary label to each annotated sample for the character attribution task. We also need to drop the very ambiguous samples and those with insufficient reliable ratings to ensure high quality.

The MTurk workers answer *yes*, *maybe yes*, *not sure*, *maybe no* or *no* on each question. We map this ordinal scale to a numeric range by mapping the labels to 2, 1, 0, -1 and -2, respectively. Each character-trope pair is annotated by at most three reliable workers. We sum the label values for each sample to get an integer score s between -6 and 6. The higher the absolute score, the greater is the agreement among the raters. We drop samples whose absolute integer score falls below 3. For the remaining samples, we obtain the annotation confidence w by normalizing the integer score s between 0 and 1: $w = (|s| - 2)/4$. The confidence score takes values 0.25, 0.5, 0.75 and 1. Attribution models can use these scores to weigh their evaluation metrics. The sample gets the label 1 (character portrays the trope) if $s > 0$, else 0 (character does not portray the trope). We include the annotations of the individual raters in CHATTEREVAL to encourage multiannotator modeling.

Dataset	Characters	Tropes	Movies	Samples
CHATTER	2998	12967	660	88124
CHATTEREVAL	271	896	78	1061

Table 2: CHATTER and CHATTEREVAL's sizes

	Min	Max	Avg	95%tile
Movies/Character	1	5	1.04	1.0
Tropes/Character	1	1107	29.40	91.0
Words/Script	3051	87738	24614.98	34011.35
Tokens/Script	5149	158038	41952.05	58933.09
Words/Segment	3	28423	2981.28	9983.8
Tokens/Segment	4	43742	4350.89	14317.15

Table 3: CHATTER and CHATTEREVAL's statistics

4 Data Statistics

CHATTER contains 88124 character-trope pairs with an almost 50-50 split between positive and negative pairs. It covers 2998 unique characters from 660 movies and 12967 tropes. CHATTEREVAL contains 1061 human-annotated character-trope pairs, covering 271 characters, 896 tropes, and 78 movies. It contains 555 positive and 506 negative pairs. Tables 2 and 3 show some data statistics of CHATTER and CHATTEREVAL.

Most characters appear in a single movie script in our dataset. CHATTER contains attribution labels for about 30 tropes for each character. Segments indicate the portions of a movie script where a character speaks or is mentioned. As shown in Table 3, the maximum size of a movie script or a character segment can exceed 87K and 28K words, respectively. This corresponds to 158K and 43K tokens, respectively (We used the Llama3 (Dubey et al., 2024) tiktoken-based BPE tokenizer). Therefore, we must use a model capable of handling long contexts for the attribution task.

Most movies in our dataset have been produced

Character	Movies	Trope	Label
Benoit Blanc	*Knives Out (2019), Glass Onion (2022)*	The **NiceGuy** trope describes a character who is kind, friendly, morally good, and socially pleasant.	1
Mark Watney	*The Martian (2015)*	The **EarnYourHappyEnding** trope involves characters enduring significant hardship, anguish, and grief, but ultimately achieving a happy ending through hard work or love	1
Arthur Fleck	*Joker (2019)*	The **TheDogBitesBack** trope occurs when a villain is attacked or betrayed by an abused subordinate or victim who seizes the chance for revenge	0
Dr. King Schultz	*Django Unchained (2012)*	The **SoreLoser** trope describes a character who reacts to defeat with anger, accusations, and bad behavior	0

Table 4: Character-Trope examples from the CHATTEREVAL dataset

in the UK or the US after 1980. They cover a wide range of genres. The top five genres – *Action*, *Drama*, *Thriller*, *Adventure*, and *Comedy* – cover more than 50% of the movies. Table 4 shows qualitative examples from the CHATTEREVAL dataset. The tropes can relate to the character's attitude and personality (*NiceGuy*), some experience or incident (*SoreLoser*), or their overall story (*EarnYourHappyEnding*). The character attribution task entails that the model understands the trope definition, reads the scripts of the movies where the character appears, and, based on that, decides whether the character portrays the trope. Appendix D lists the most commonly occurring tropes and their definitions to visualize the trope space.

Our dataset does not contain scripts for all the movies where the character appears. For example, the character Arthur Fleck "Joker" has appeared in multiple movies, comics, and TV shows, but CHATTER only contains the movie script of the *Joker (2019)* movie. The TVTropes contributors and our raters draw their knowledge of the character from all sources. However, the attribution model predicts the attribute labels based only on the screenplay and its pretraining data. Therefore, intractable character-trope pairs could exist for which the model has insufficient information. In future work, we will investigate ways to find such intractable samples in our dataset.

5 Experiments

5.1 Models

We establish baselines on CHATTEREVAL using zero-shot and few-shot prompting. We used two closed-source models, Gemini-1.5-Flash (Reid et al., 2024) and GPT-4o-mini (Hurst et al., 2024), and three open-source models, Phi-3-small-7B-128k-Instruct (Abdin et al., 2024), Llama-3.1-8B-Instruct (Dubey et al., 2024) and Mistral-Nemo-Instruct-12B-2407 (Jiang et al., 2023), in our experiments. We selected these models because they can handle long contexts (128K tokens).

We experiment with four prompting strategies. **1) Priors** - We prompt the model with the character name, the list of movies where the character has appeared, and the trope definition. We do not include any screenplay content and ask the model to find the attribution label based solely on its prior knowledge about the character. **2) Script** - We include the full movie script in the prompt. **3) Segment (Zero-shot)** - We include the segments of the movie script where the character speaks or is mentioned. **4) Segment (Few-shot)** - We include the character segments and two examples from the CHATTEREVAL dataset. We select examples of the same trope. We also experimented with selecting random examples or examples of the same character, but they both performed worse than selecting same-trope examples. We do not apply few-shot prompting with movie scripts because the prompt size becomes too large. Appendix E describes the prompt template we used.

For the last three settings – **Script**, **Segment (Zero-shot)**, and **Segment (Few-shot)** – we replaced character names in the movie scripts or the character segments with random alphanumeric identifiers. We hypothesize that the LLM could be generating its response by recognizing the character name and using the knowledge it had accrued about the character from its pretraining datasets. The character-anonymization step possibly prevents the LLM from relying on its prior knowledge about the character and forces it to decide the attribution label based solely on the given movie script or character segments. We observed that prompting with the non-anonymized documents produced better performance, validating our hypothesis. Future work should explore more effective prompting strategies to nullify the effect of the model's priors

Prompt	Model	Acc	P	R	F1
	Random	50.0	52.3	50.0	51.1
	CHATTER	*80.6*	*81.0*	*82.2*	*81.6*
Priors	Gemini-1.5	**81.9**	**91.4**	**72.2**	**80.7**
	GPT-4o	76.2	98.1	55.6	71.0
	Phi-7B	56.6	89.1	19.5	32.0
	Llama-8B	71.0	68.9	81.4	74.6
	Mistral-12B	73.1	83.0	61.3	70.5
Script	Gemini-1.5	72.4	83.1	59.4	69.3
	GPT-4o	**73.9**	**71.2**	**84.4**	**77.2**
	Phi-7B	65.5	77.4	48.1	59.3
	Llama-8B	61.4	61.4	70.7	65.7
	Mistral-12B	56.1	60.5	45.7	52.1
Segment (Zero-Shot)	Gemini-1.5	73.5	86.6	58.4	69.8
	GPT-4o	**78.5**	**77.4**	**83.3**	**80.3**
	Phi-7B	73.3	77.1	68.8	72.7
	Llama-8B	69.8	65.6	87.7	75.1
	Mistral-12B	69.3	81.4	53.2	64.3
Segment (Few-Shot)	Gemini-1.5	65.5	93.4	36.7	52.7
	GPT-4o	**74.5**	**79.5**	**69.2**	**74.0**
	Phi-7B	58.4	58.3	93.2	71.7
	Llama-8B	60.8	61.0	75.3	67.4
	Mistral-12B	62.0	62.0	73.6	67.3

Table 5: Prior, Zero-shot, and Few-shot performance on CHATTEREVAL for the character attribution binary classification task. The **CHATTER** row uses the CHATTER's labels as predictions. We bolden the model row with the best accuracy (or F1) for each prompting strategy.

on the generated response.

We perform greedy decoding. We also evaluate CHATTER's labels against the annotations of CHATTEREVAL to assess its suitability as a training set for character attribution. We use permutation tests at $\alpha = 0.05$ to compare the performance of different prompting strategies and models. We apply Bonferroni correction to correct for multiple comparisons.

5.2 Results

Table 5 shows the performance of the different prompting strategies. The closed-sourced models were significantly more accurate than the open-sourced models for all strategies. This is expected because the closed-sourced models supposedly contain more parameters and have been trained on more extensive data. Gemini-1.5 has the strongest prior knowledge, followed by GPT-4o. They performed equally well when we prompted them on the full movie script. However, GPT-4o's zero-shot and few-shot accuracy on character segments was significantly better than Gemini's. There was no significant difference between the accuracy scores of Phi-7B, Llama-8B, and Mistral-12B for any of the prompting strategies, except for priors where Phi-7B performed worse than random.

Comparing the different prompting strategies, we observed that the zero-shot accuracy on character segments was significantly better than the few-shot accuracy across all the models. This discrepancy in performance is surprising because prompting models usually provide better results with exemplars in the prompt. A possible reason could be that the two examples in the prompt are insufficient to represent the task adequately. Increasing the number of prompts is not viable because of the context limit. A possible solution could be summarizing the character segments to fit more examples in the prompt.

Prompting the character segments usually performed as well or better than prompting the full movie scripts. However, in most cases, the model's prior performance already showed strong results. This confirms that these models were probably pretrained on the tropes data from TVTropes, making it harder to evaluate the true character attribution capability of the model. The huge gap in performance between the priors of the closed-source models and the zero-shot or few-shot prompting results of the open-sourced models shows that there is much scope for improvement for the open-sourced models. This is important because, given a movie script (or any other narrative document), we do not always want to prompt it using commercial APIs because they might contain private or unpublished information. Training local attribution models on the character-trope pairs of CHATTER should narrow this performance gap.

There is no significant difference between the results of the strongest-performing model and using the labels of the CHATTER dataset as predictions. Therefore, CHATTER can serve as a good training source for the character attribution task. The strong zero-shot performance of the closed-sourced models suggests that we can use them to create good-quality synthetic training data.

We also observe that the recall scores of the models are usually lower than their precision. A possible reason could be the sizeable prompt size, which makes it difficult for the model to pinpoint the relevant sections from the script or the character segments. While training character attribution models, we should further preprocess the data for more effective learning.